

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 16

DATE : 02-01-2020

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 16

Topics

DATE : 02 - 01 - 2020

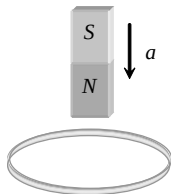
PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]**PHYSICS** : Wave Optics, Modern Physics**CHEMISTRY** : Electrochemistry, Mole + Concentration Unit + Solution**BOTANY** : Plant Anatomy(Sec. Growth),Transport in plants,Microbe in Human Welfare,
Plant Breeding

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. Lenz's law is consequence of the law of conservation of :
 (1) Charge
 (2) Momentum
 (3) Mass
 (4) Energy
2. A metallic ring is attached with the wall of a room. When the north pole of a magnet is brought near to it, the induced current in the ring will be

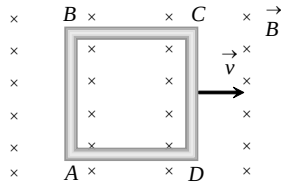


- (1) First clockwise then anticlockwise
 - (2) In clockwise direction
 - (3) In anticlockwise direction
 - (4) First anticlockwise then clockwise
3. The magnetic flux linked with a coil is given by an equation ϕ (in webers) = $8t^2 + 3t + 5$. The induced e.m.f. in the coil at the fourth second will be
 (1) 16 units
 (2) 39 units
 (3) 67 units
 (4) 145 units
 4. A coil having an area 2m^2 is placed in a magnetic field which changes from 1 Wb/m^2 to 4 Wb/m^2 in a interval of 2 second. The e.m.f. induced in the coil will be
 (1) 4 V
 (2) 3 V
 (3) 1.5 V
 (4) 2 V

5. The magnetic field in a coil of 100 turns and 40 square cm area is increased from 1 Tesla to 6 Tesla in 2 second. The magnetic field is perpendicular to the coil. The e.m.f. generated in it is :
 (1) 10^4 V
 (2) 1.2 V
 (3) 1.0 V
 (4) 10^{-2} V
6. A 10 metre wire kept in east-west falling with velocity 5 m/sec perpendicular to the field $0.3 \times 10^{-4}\text{ Wb/m}^2$. The induced e.m.f. across the terminal will be :
 (1) 0.15 V
 (2) 1.5 mV
 (3) 1.5 V
 (4) 15.0 V
7. A conductor of 3 m in length is moving perpendicularly to magnetic field of 10^{-3} tesla with the speed of 10^2 m/s , then the e.m.f. produced across the ends of conductor will be :
 (1) 0.03 volt
 (2) 0.3 volt
 (3) 3×10^{-3} volt
 (4) 3 volt

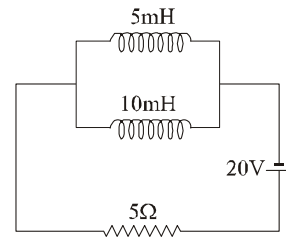
SPACE FOR ROUGH WORK

8. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane of the loop exists everywhere. The current induced in the loop is :



- (1) $\frac{Blv}{R}$ clockwise
(2) $\frac{Blv}{R}$ anticlockwise
(3) $\frac{2Blv}{R}$ anticlockwise
(4) Zero
9. A wheel with ten metallic spokes each 0.50 m long is rotated with a speed of 120 rev/min in a plane normal to the earth's magnetic field at the place. If the magnitude of the field is 0.4 Gauss, the induced e.m.f. between the axle and the rim of the wheel is equal to
(1) $1.256 \times 10^{-3} \text{ V}$
(2) $6.28 \times 10^{-4} \text{ V}$
(3) $1.256 \times 10^{-4} \text{ V}$
(4) $6.28 \times 10^{-5} \text{ V}$

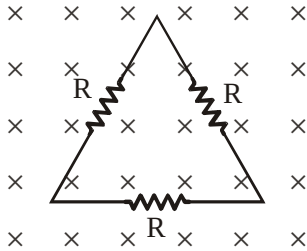
10. A rod of length 20 cm is rotating with angular speed of 100 rps in a magnetic field of strength 0.5 T about its one end. What is the potential difference between two ends of the rod
(1) 2.28 V (2) 4.28 V
(3) 6.28 V (4) 2.5 V
11. In the given circuit, the current through 5mH inductor in steady state is



- (1) $4/3$ Amp
(2) $8/3$ Amp
(3) 4 Amp
(4) $2/3$ Amp
12. In which of the following cases the emf is induced due to time varying magnetic field :-
Case - I : A magnet is moving along the axis of conducting coil.
Case - II : A loop having varying area and is placed in uniform magnetic field.
Case - III : A current carrying wire lying in the plane of the loop is approaching a closed loop.
(1) I, II and III
(2) Only I & II
(3) Only I & III
(4) Only I

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13. Three resistance of magnitude R each are connected in the form of an equilateral triangle of side a . The combination is placed in magnetic field $B = B_0 e^{-\lambda t}$ perpendicular to the plane. The induced current in the circuit is given by :-



- (1) $\left(\frac{a^2 \lambda}{2\sqrt{3}R} B_0 \right) e^{-\lambda t}$ (2) $\left(\frac{a^2 \lambda}{4\sqrt{3}R} B_0 \right) e^{-\lambda t}$
 (3) $\left(\frac{a^2 B_0}{4\lambda\sqrt{3}R} \right) e^{-\lambda t}$ (4) $\left(\frac{a^2 B_0 R}{4\lambda\sqrt{3}} \right) e^{-\lambda t}$
14. A circuit area 0.01 m^2 is kept inside a magnetic field which is normal to its plane. The magnetic field changes from 2 T to 1 T in 1 millisecond . If the resistance of the circuit is 2Ω . The amount of heat evolved is :-
 (1) 0.05 J (2) 50 J
 (3) 0.50 J (4) 500 J
15. A rod of length ℓ rotates with a uniform angular velocity ω about an axis passing through its middle point but normal to its length in a uniform magnetic field of induction B with its direction parallel to the axis of rotation. The induced emf between the two ends of the rods is :-

- (1) $\frac{B\ell^2\omega}{2}$ (2) zero
 (3) $(B\ell^2\omega/8)$ (4) $2B\ell^2\omega$

16. When a metallic plate swings between the poles of a magnet :-

- (1) no effect on the velocity of plate
 (2) eddy currents are set up inside the plate and the direction of the current increases the velocity
 (3) eddy currents are set up inside the plate and the direction of the current oppose the motion of the plate
 (4) eddy currents cannot set up inside the plate.

17. What is the charge induced in coil of 100 turns of resistance 100Ω , if magnetic flux changes from 2 Tm^2 to -2 Tm^2 :-

- (1) 4 C (2) 2 C
 (3) 2.8 C (4) 0.4 C

18. In an A.C. circuit V and I are given by

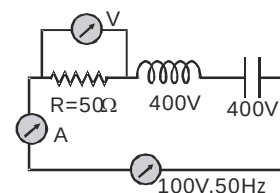
$$V = 100 \sin(100 t) \text{ volts and}$$

$$I = 100 \sin\left[100t + \frac{\pi}{3}\right] \text{ mA}$$

The power dissipated in the circuit is :-

- (1) 10^4 watt (2) 10 watt
 (3) 2.5 watt (4) 5.0 watt

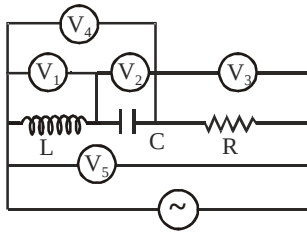
19. In given LCR circuit, the voltage across the terminals of a resistance and current will be :-



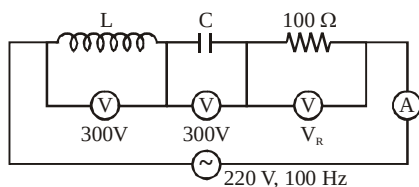
- (1) $400\text{V}, 2\text{A}$ (2) $800\text{V}, 2\text{A}$
 (3) $100\text{V}, 2\text{A}$ (4) $100\text{V}, 4\text{A}$

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20. In the adjoining A.C. circuit the voltmeter whose reading will be zero at resonance is :-



- (1) V_1 (2) V_2
(3) V_3 (4) V_4
21. A series R-L-C circuit consists of an 8.0Ω resistor, a $5.0 \mu\text{F}$ capacitor and a 50.0 mH inductor. A variable frequency source applies an emf of 400 V (rms) across the combination. The power delivered to the circuit when the frequency is equal to one-half the resonance frequency is :-
- (1) 52 W (2) 57 W
(3) 63 W (4) 69 W
22. In an LCR circuit shown in the following figure, what will be the readings of the voltmeter across the resistor and ammeter if an a.c. source of 220 V and 100 Hz is connected to it as shown ?

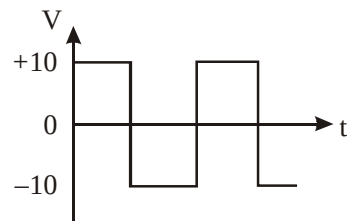


- (1) $300 \text{ V}, 3 \text{ A}$
(2) $800 \text{ V}, 8 \text{ A}$
(3) $110 \text{ V}, 1.1 \text{ A}$
(4) $220 \text{ V}, 2.2 \text{ A}$

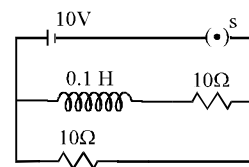
23. The network shown in the figure is a part of a complete circuit. If at a certain instant the current i is 5 A and is decreasing at the rate of 10^3 A/s then $V_A - V_B$ is :-



- (1) 5 V (2) 10 V
(3) 15 V (4) 20 V
24. The r.m.s. voltage of the wave form shown is :



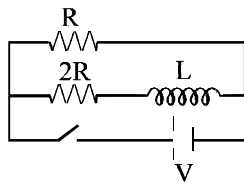
- (1) 10 V
(2) 7 V
(3) 6.37 V
(4) None of these
25. In the adjoining circuit, initially the switch S is open. The switch ' S ' is closed at $t = 0$. The difference between the maximum and minimum current that can flow in the circuit is



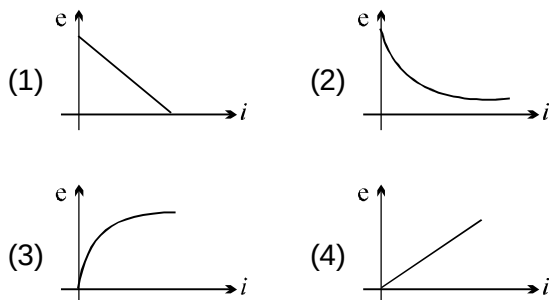
- (1) 2 Amp
(2) 3 Amp
(3) 1 Amp
(4) nothing can be concluded

SPACE FOR ROUGH WORK

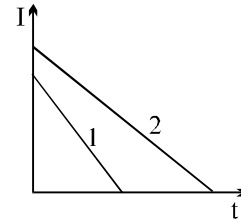
26. The ratio of time constant in build up and decay in the circuit shown in figure is



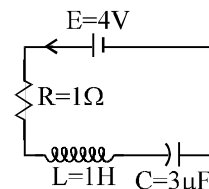
- (1) 1 : 1 (2) 3 : 2
(3) 2 : 3 (4) 1 : 3
27. In an L-R circuit connected to a battery of constant e.m.f. E switch S is closed at time $t = 0$. If e denotes the magnitude of induced e.m.f. across inductor and i the current in the circuit at any time t . Then which of the following graphs shows the variation of e with i ?



28. A current of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H. The energy stored in the inductor per unit time is
(1) 2 J/s (2) 1 J/s
(3) 16 J/s (4) 4 J/s
29. Two identical inductance carry currents that vary with time according to linear laws (as shown in figure). In which of two inductance is the self induction emf greater?



- (1) 1
(2) 2
(3) same
(4) data are insufficient to decide
30. The current in the given circuit is increasing with a rate $a = 4$ amp/s. The charge on the capacitor at an instant when the current in the circuit is 2 amp will be :



- (1) $4\mu\text{C}$ (2) $5\mu\text{C}$
(3) $6\mu\text{C}$ (4) none of these
31. L , C and R represent physical quantities inductance, capacitance and resistance. The combination which has the dimensions of frequency is
(1) $\frac{1}{RC}$ and $\frac{R}{L}$
(2) $\frac{1}{\sqrt{RC}}$ and $\sqrt{\frac{R}{L}}$
(3) $\sqrt{LC}$
(4) $\frac{C}{L}$

SPACE FOR ROUGH WORK

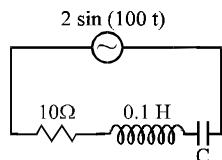
32. A coil of inductance 5H is joined to a cell of emf 6V through a resistance 10Ω at time $t = 0$. The emf across the coil at time $t = \ln \sqrt{2}$ s is:

(1) 3V (2) 1.5V
(3) 0.75V (4) 4.5V

33. A long solenoid of N turns has a self inductance L and area of cross section A . When a current i flows through the solenoid, the magnetic field inside it has magnitude B . The current i is equal to:

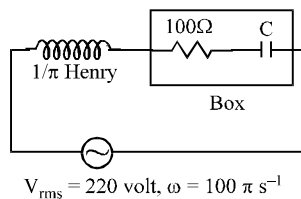
(1) BAN/L (2) $BANL$
(3) BN/AL (4) B/ANL

34. The power factor of the circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to



(1) $400\ \mu\text{F}$ (2) $300\ \mu\text{F}$
(3) $500\ \mu\text{F}$ (4) $200\ \mu\text{F}$

35. In the circuit, as shown in the figure, if the value of R.M.S current is 2.2 ampere, the power factor of the box is



(1) $\frac{1}{\sqrt{2}}$ (2) 1
(3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

36. When 100V DC is applied across a solenoid a current of 1A flows in it. When 100V AC is applied across the same coil, the current drops to 0.5A . If the frequency of the AC source is 50Hz , the impedance and inductance of the solenoid are:

(1) $100\Omega, 0.93\text{H}$ (2) $200\Omega, 1.0\text{H}$
(3) $10\Omega, 0.86\text{H}$ (4) $200\Omega, 0.55\text{H}$

37. In ac circuit when ac ammeter is connected it reads i current if a student uses dc ammeter in place of ac ammeter the reading in the dc ammeter will be:

(1) $\frac{i}{\sqrt{2}}$ (2) $\sqrt{2}i$

(3) $0.637i$ (4) zero

38. The phase difference between current and voltage in an AC circuit is $\pi/4$ radian. If the frequency of AC is 50Hz , then the phase difference is equivalent to the time difference :

(1) 0.78s (2) 15.7ms
(3) 0.25s (4) 2.5ms

39. Power factor of an L-R series circuit is 0.6 and that of a C-R series circuit is 0.5 . If the element (L, C, and R) of the two circuits are joined in series the power factor of this circuit is found to be 1 . The ratio of the resistance in the L-R circuit to the resistance in the C-R circuit is

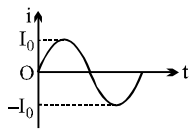
(1) $6/5$ (2) $5/6$
(3) $\frac{4}{3\sqrt{3}}$ (4) $\frac{3\sqrt{3}}{4}$

SPACE FOR ROUGH WORK

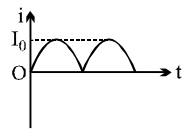
40. The effective value of current $i = 2 \sin 100 \pi t + 2 \sin(100 \pi t + 30^\circ)$ is :

- (1) $\sqrt{2}$ A (2) $2\sqrt{2+\sqrt{3}}$
(3) 4 (4) None

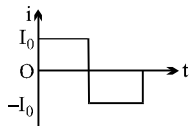
41. If I_1 , I_2 , I_3 and I_4 are the respective r.m.s. values of the time varying currents as shown in the four cases I, II, III and IV. Then identify the correct relations.



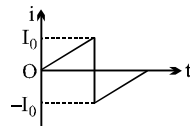
I



II



III



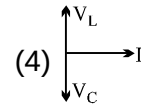
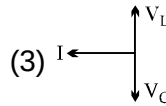
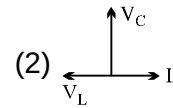
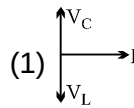
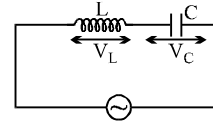
IV

- (1) $I_1 = I_2 = I_3 = I_4$ (2) $I_3 > I_1 = I_2 > I_4$
(3) $I_3 > I_4 > I_2 = I_1$ (4) $I_3 > I_2 > I_1 > I_4$

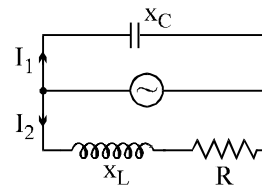
42. In series LR circuit $X_L = 3R$. Now a capacitor with $X_C = R$ is added in series. Ratio of new to old power factor is

- (1) 1 (2) 2 (3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{2}$

43. The current I , potential difference V_L across the inductor and potential difference V_C across the capacitor in circuit as shown in the figure are best represented vectorially as



44. In the shown AC circuit phase different between currents I_1 and I_2 is



(1) $\frac{\pi}{2} - \tan^{-1} \frac{X_L}{R}$

(2) $\tan^{-1} \frac{X_L - X_C}{R}$

(3) $\frac{\pi}{2} + \tan^{-1} \frac{X_L}{R}$

(4) $\tan^{-1} \frac{X_L - X_C}{R} + \frac{\pi}{2}$

45. In a series R-L-C circuit, the frequency of the source is half of the resonance frequency. The nature of the circuit will be

- (1) capacitive
(2) inductive
(3) purely resistive
(4) data insufficient

SPACE FOR ROUGH WORK

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| <p>46. Density of water is 1g/mL. The concentration of water in mol/litre is-</p> <p>(1) 1000 (2) 18
(3) 0.018 (4) 55.5</p> <p>47. 1000 gram aqueous solution of CaCO_3 contains 10 gram of CaCO_3. Concentration of solution is-</p> <p>(1) 10 ppm (2) 100 ppm
(3) 1000 ppm (4) 10,000 ppm</p> <p>48. Equal volumes of 0.1 M AgNO_3 and 0.2 M NaCl are mixed. The concentration of NO_3^- ions in the mixture will be-</p> <p>(1) 0.1 M (2) 0.05 M
(3) 0.2 M (4) 0.15 M</p> <p>49. 10 gram of glucose are dissolved in 150 gram of water. The mass % of glucose is-</p> <p>(1) 5% (2) 6.25%
(3) 93.75% (4) 15%</p> <p>50. Vapour pressure of a solution containing nonvolatile solute is -</p> <p>(1) More than the vapour pressure of a solvent
(2) Less than the vapour pressure of solvent
(3) Equal to the vapour pressure of solvent
(4) None of these</p> <p>51. In cold countries, ethylene glycol is added to water in the radiators of cars during winters. It results in-</p> <p>(1) Lowering in boiling point
(2) Reducing viscosity
(3) Reducing specific heat
(4) Lowering in freezing point</p> | <p>52. An example of colligative property is-</p> <p>(1) Freezing point
(2) Boiling point
(3) Vapour pressure
(4) Osmotic pressure</p> <p>53. Which solution will show maximum elevation in b.pt-</p> <p>(1) 0.1 M KCl
(2) 0.1 M BaCl_2
(3) 0.1 M FeCl_3
(4) 0.1 M $\text{Fe}_2(\text{SO}_4)_3$</p> <p>54. 0.5 M solution of urea is isotonic with-</p> <p>(1) 0.5 M NaCl solution
(2) 0.5 M sugar solution
(3) 0.5 M BaCl_2 solution
(4) 0.5 M solution benzoic acid in benzene</p> <p>55. Which one of the following salts would have the same value of the Vant Hoff factor (i) as that of $\text{K}_3[\text{Fe}(\text{CN})_6]$.</p> <p>(1) NaCl (2) Na_2SO_4
(3) $\text{Al}_2(\text{SO}_4)_3$ (4) $\text{Al}(\text{NO}_3)_3$</p> <p>56. Decimolar solution of potassium ferricyanide, $\text{K}_3[\text{Fe}(\text{CN})_6]$ has osmotic pressure of 3.94 atm at 27°C. Hence percent ionisation of the solute is -</p> <p>(1) 10% (2) 20% (3) 30% (4) 40%</p> <p>57. The osmotic pressure of a solution of benzoic acid dissolved in benzene is less than expected because -</p> <p>(1) Benzoic acid is an organic solute
(2) Benzene is a non-polar solvent
(3) Benzoic acid dissociates in benzene
(4) Benzoic acid gets associated in benzene</p> |
|---|--|

SPACE FOR ROUGH WORK

- | | |
|---|---|
| <p>58. pH of 1M HA (weak acid) is 2. Hence van't Hoff factor is -
 (1) 1.2 (2) 1.02
 (3) 1.1 (4) 1.01</p> <p>59. Two liquids A and B form an ideal solution. At 300 K the vapour pressure of a solution of 1 mole of A and x moles of B is 550 mm. If the vapour pressures of pure A and B are 400 mm and 600 mm respectively, then x is -
 (1) 1 (2) 2 (3) 3 (4) 4</p> <p>60. Blood cells retain their normal shapes in solutions which are-
 (1) isotonic to blood
 (2) hypotonic to blood
 (3) hypertonic to blood
 (4) all of these</p> <p>61. Mass of 1 atom of Hydrogen is -
 (1) 1.66×10^{-24} g (2) 10^{-22} g
 (3) 10^{-23} g (4) 10^{-25} g</p> <p>62. What is correct for 10 g of CaCO_3 -
 (1) It contains 1g-atom of carbon
 (2) It contains 0.3 g-atoms of oxygen
 (3) It contains 12 g of calcium
 (4) None of these</p> <p>63. No. of oxalic acid molecules in 100 ml of 0.02 N oxalic acid is -
 (1) 6.023×10^{20} (2) 6.023×10^{21}
 (3) 6.023×10^{22} (4) 6.023×10^{23}</p> <p>64. Equal masses of O_2, H_2 and CH_4 are taken in a container. The respective mole ratio of these gases in container is -
 (1) 1 : 16 : 2 (2) 16 : 1 : 2
 (3) 1 : 2 : 16 (4) 16 : 2 : 1</p> | <p>65. The mass percent of N in 66% pure $(\text{NH}_4)_2\text{SO}_4$ sample is -
 (1) 32 (2) 28
 (3) 14 (4) None of these</p> <p>66. 14g of element X combine with 16g of oxygen. On the basis of this information, which of the following is a correct statement :
 (1) The element X could have an atomic weight of 7 and its oxide formula XO
 (2) The element X could have an atomic weight of 14 and its oxide the formula X_2O
 (3) The element X could have an atomic weight of 7 and its oxide is X_2O
 (4) The element X could have an atomic weight of 14 and its oxide is XO_2</p> <p>67. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $\text{Ca}(\text{OH})_2$. The maximum number of moles of CaSO_4 formed is -
 (1) 0.2 (2) 0.5
 (3) 0.4 (4) 1.5</p> <p>68. The law of conservation of mass holds good for all of the following except -
 (1) All chemical reactions
 (2) Nuclear reactions
 (3) Endothermic reactions.
 (4) Exothermic reactions.</p> <p>69. The vapour density of gas A is four times that of B. If molecular mass of B is M, then molecular mass of A is -
 (1) M (2) 4M (3) $\frac{M}{4}$ (4) 2M</p> |
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SPACE FOR ROUGH WORK

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| <p>70. If LPG cylinder contains mixture of butane and isobutane, then the amount of oxygen that would be required for combustion of 1kg of it will be-</p> <p>(1) 1.8 kg (2) 2.7 kg
(3) 4.5 kg (4) 3.58 kg</p> <p>71. 8 gm of O_2 has the same number of molecules as-</p> <p>(1) 7 gm of CO
(2) 14 gm of CO
(3) 14 gm of CO_2
(4) 12 gm of CO_2</p> <p>72. A sample of AlF_3 contains 3.0×10^{24} F^- ions. The number of formula units in this sample are -</p> <p>(1) 9.0×10^{24}
(2) 3.0×10^{24}
(3) 0.75×10^{24}
(4) 1.0×10^{24}</p> <p>73. A sample of hard water is found to contain 40 mg of Ca^{+2} ion per litre. The amount of washing soda (Na_2CO_3) required to soften five litres of the sample would be -</p> <p>(1) 1.06 gm (2) 5.3 gm
(3) 53 mg (4) 530 mg</p> <p>74. 1.5 gm of divalent metal displaced 4 gm of copper (at. wt. = 64) from a solution of copper sulphate. The atomic weight of the metal is-</p> <p>(1) 12 (2) 24
(3) 48 (4) 6</p> | <p>75. Avogadro's number of Rupees can be spent inyears if 10 lac rupees per second are spent -</p> <p>(1) 1.91×10^{10} year
(2) 2.91×10^{10} year
(3) 3.91×10^{10} year
(4) 4.91×10^{10} year</p> <p>76. How much will the reduction potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralized to pH = 7 ?</p> <p>(1) Increase by 0.059 V
(2) Decrease by 0.059 V
(3) Increase by 0.41 V
(4) Decrease by 0.41 V</p> <p>77. The charge required for the oxidation of one mole Mn_3O_4 into MnO_4^{2-} is</p> <p>(1) 5×96500 C
(2) 96500 C
(3) 10×96500 C
(4) 2×96500 C</p> <p>78. In the electrolysis of aqueous solution of NaOH, 2.8 litre of oxygen was liberated at the anode at NTP. How much hydrogen was liberated at the cathode ?</p> <p>(1) 5.6 litre (2) 56 ml
(3) 560 ml (4) 0.056 L</p> <p>79. The standard electrode potential of $Cu Cu^{+2}$ is -0.34 Volt. At what concentration of Cu^{+2} ions, electrode potential will be zero ?</p> <p>(1) 1.98×10^{-12} M
(2) 2.98×10^{-12} M
(3) 1.49×10^{-12} M
(4) 5.96×10^{-12} M</p> |
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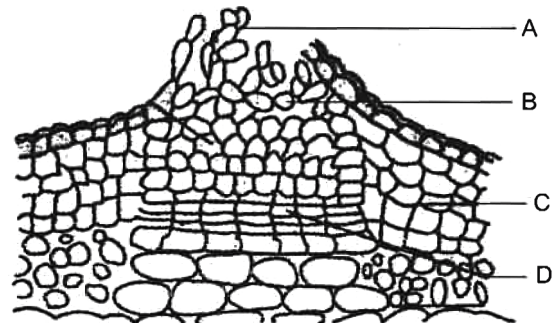
80. For which of the following electrolytes, molar conductance will be three times that of equivalence conductance at any dilution?
 (1) $\text{Al}_2(\text{SO}_4)_3$ (2) Na_3PO_4
 (3) MgCl_2 (4) NaCl
81. Equivalent conductance at infinite dilution of BaCl_2 , H_2SO_4 , and HCl solutions in water are x , y and z respectively. Equivalent conductance of BaSO_4 solution is :
 (1) $x - y + z$ (2) $x + y - 2z$
 (3) $x + 2y + z$ (4) $x - y + 2z$
82. The strength of current in amp will required to liberate 0.28L Cl_2 at S.T.P from sodium chloride solution in 965 second is
 (1) 2.0 amp (2) 2.5 amp
 (3) 1.0 amp (4) 1.5 amp
83. $\text{Mg}_{(s)} | \text{Mg}^{+2} (0.1 \text{ M}) || \text{Ag}^+ (1 \times 10^{-4} \text{ M}) | \text{Ag}$
 $E^\circ_{\text{Ag}^+|\text{Ag}} = 0.8 \text{ V}$, $E^\circ_{\text{Mg}^{+2}|\text{Mg}} = -2.37 \text{ V}$
 The emf of above cell is
 (1) 3.17 V (2) 3.38 V
 (3) 2.79 V (4) 2.96 V
84. $\text{Pt}, \text{H}_2 | \text{HCl} | \text{H}_2, \text{Pt}$
 4 atm 2 atm
 The emf of above cell is
 (1) $8.9 \times 10^{-3} \text{ V}$
 (2) $-8.9 \times 10^{-3} \text{ V}$
 (3) $4.45 \times 10^{-2} \text{ V}$
 (4) $-4.45 \times 10^{-2} \text{ V}$
85. $\text{Cu}^{+2}_{(aq)} + \text{Sn}^{+2}_{(aq)} \rightarrow \text{Sn}^{+4}_{(aq)} + \text{Cu}_{(s)}$
 $E^\circ_{\text{Cu}^{+2}|\text{Cu}} = 0.34 \text{ V}$ $E^\circ_{\text{Sn}^{+4}|\text{Sn}^{+2}} = 0.15 \text{ V}$
 The equilibrium constant of above reaction is (Antilog 0.42 = 2.63)
 (1) 2.63×10^7
 (2) 2.63×10^5
 (3) 2.63×10^4
 (4) 2.63×10^6
86. $\text{Pt}, (\text{Cl}_2) | \text{Cl}^-(\text{C}_2) || \text{Cl}^-(\text{C}_1) | \text{Pt}, (\text{Cl}_2)$
 1 atm (Anode) 1 atm (Cathode)
 The above cell reaction is spontaneous if
 (1) $\text{C}_2 > \text{C}_1$ (2) $\text{C}_2 < \text{C}_1$
 (3) $\text{C}_2 = \text{C}_1$ (4) $\text{C}_2 > 0$
87. If $E^\circ_{\text{Fe}^{+2}|\text{Fe}} = x_1$ and $E^\circ_{\text{Fe}^{+3}|\text{Fe}} = x_2$ then $E^\circ_{\text{Fe}^{+3}|\text{Fe}^{+2}}$ will be
 (1) $x_1 + x_2$ (2) $3x_2 + 2x_1$
 (3) $3x_2 - 2x_1$ (4) $x_2 - x_1$
88. At cathode, which cation reduce first?
 (1) Mg^{+2} (2) Zn^{+2}
 (3) Cr^{+3} (4) Ag^+
89. When a lead storage battery is discharged
 (1) SO_2 is evolved
 (2) Lead sulphate is consumed
 (3) Lead is formed
 (4) Sulphuric acid is consumed
90. Saturated solution of KNO_3 is used to make salt bridge because
 (1) Velocity of K^+ is greater than that of NO_3^-
 (2) Velocity of NO_3^- is greater than that of K^+
 (3) Velocity of both K^+ and NO_3^- are nearly the same
 (4) KNO_3 is highly soluble in water

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91. Which of the following type of culture is used, in some interspecific crosses where endosperm of developing hybrid seed degenerates very early :-
 (1) Meristem culture
 (2) Shoot tip culture
 (3) Embryo culture
 (4) Anther culture
92. Pomato is the product of :-
 (1) r-DNA technique
 (2) Mutation
 (3) Somatic hybridization
 (4) Selection
93. Embryoids is -
 (1) Highly active embryo
 (2) Embryo produced from reproductive cells.
 (3) Embryo produced from somatic cell in tissue culture
 (4) Dead embryo
94. Regeneration from callus or protoplast leads to the generation of considerable variation, described as-
 (1) Somaclonal variation
 (2) Biological variation
 (3) Micro propagation
 (4) Clonal propagation
95. In thistle funnel experiment 5 Pa pressure is required to prevent the entry of water into funnel, then what is the value of osmotic potential ?
 (1) To calculate the value of osmotic potential, value of ψ_w is required
 (2) -5 Pa
 (3) 5 Pa
 (4) There is no osmotic potential in the system
96. Organic and mineral nutrients undergo _____ transport.
 (1) Multidirectional
 (2) Bidirectional
 (3) Unidirectional
 (4) Non directional
97. At atmospheric pressure solution 'A' has water potential of -1000 Pa and solution 'B' has water potential of -500 Pa. Which solution has lower solute potential ?
 (1) Solution B
 (2) Both the solutions have same solute potential
 (3) Solution A
 (4) Solute potential cannot be calculated
98. The porins are proteins that form huge pores in the outer membranes of the :-
 (1) Some bacteria
 (2) Plastids
 (3) Mitochondria
 (4) All of the above
99. Most of the herbaceous plants have about _____ water in their fresh weight.
 (1) 10 to 15 percent
 (2) 92 percent
 (3) 85 to 90 percent
 (4) 8 percent
100. Cells swell in the solutions those have _____ concentration of solutes and shrink in the solutions those have _____ concentration of solutes as compare to cells.
 (1) More, less (2) Equal, more
 (3) Less, more (4) Less, equal
101. In plant cells, which of the following are important determinants of movement of molecules in or out of the cell ?
 (1) Cell wall and cell membrane
 (2) Cell wall and tonoplast
 (3) Cell wall and glycocalyx
 (4) Cell membrane and the membrane of the vacuole
102. Cell A with $\psi_s = -20$, TP = 10 and cell B with TP = 15 are in contact. If we want the net direction of water movement from cell A to B, then value of ψ_s in cell B should be :-
 (1) Higher than cell A
 (2) Lower than cell A
 (3) Similar to cell A
 (4) There is no effect of value of ψ_s

103. Rubber cork, placed in water, not swells up (not show imbibition) due to :-
 (1) Lack of concentration gradient
 (2) Lack of selective permeable membrane
 (3) Lack of affinity between imbibate and imbibant
 (4) Lipophobic nature
104. Interfascicular cambium develops from the cells of :-
 (1) Pericycle
 (2) Medullary rays
 (3) Xylem parenchyma
 (4) Endodermis
105. Select the correct statement from the following
 (1) All lateral meristems are secondary meristem
 (2) All secondary meristems are lateral meristem
 (3) All intercalary meristems are secondary meristem
 (4) All secondary meristems are intercalary meristem.
106. Which of the following tissues are included in bark? (According to modern view)
 (1) Secondary xylem, periderm and secondary phloem
 (2) Secondary cortex, secondary phloem and xylem.
 (3) Dermatogen, periblem, plerome and secondary phloem.
 (4) Phellem, Phellogen, Phelloderm and secondary phloem.
107. Which of the following are examples of lateral meristems
 (1) Fascicular vascular combium
 (2) Inter-fascicular cambium
 (3) Cork cambium
 (4) All of the above

108. The autumn wood is A in colour and has a B density. A and B are respectively :-
 (1) Lighter, Higher
 (2) Darker, Higher
 (3) Lighter, Lower
 (4) Darker, Lower
109. The portion of the oldest layer of secondary xylem is :
 (1) Just inside the vascular cambium
 (2) Just inside the primary xylem
 (3) Just outside the vascular cambium
 (4) Just outside the primary xylem
110. All the tissues exterior to vascular cambium are termed as
 (1) Phelloderm
 (2) Periderm
 (3) Bark
 (4) Cork
111. In the given diagram, identify A, B, C and D.



	A	B	C	D
(1)	Epidermis	Complement ary cells	Phellogen	Phelloderm
(2)	Epidermis	Phellem	Phellogen	Phelloderm
(3)	Phelloderm	Periderm	Phellogen	Phellem
(4)	Epidermis	Cork	Cork cambium	Secondary cortex

112. Alburnum is
 (1) Dark coloured
 (2) Light coloured and physiologically inactive
 (3) Simple permanent tissue
 (4) Commercially very important

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| <p>113. During secondary growth in dicot stem, cork cambium has cut off few cells which are impermeable to water. These cell have been cut off towards:</p> <ol style="list-style-type: none"> (1) Outer side (2) Inner side (3) Lateral side (4) In all directions <p>114. Differential activity of cambium is observed in :-</p> <ol style="list-style-type: none"> (1) Stem in tropical region (2) Stem in temperate region (3) Root in tropical region (4) Root in herbaceous plant <p>115. Which type of wood is incorrectly matched with its given feature?</p> <ol style="list-style-type: none"> (1) Heartwood—Provides mechanical support (2) Spring wood—Possess vessels with wider lumen (3) Autumn wood—Darker in appearance (4) Sapwood—Functionally inactive <p>116. In hardening process :-</p> <ol style="list-style-type: none"> (1) Plantlet is placed in reduced light and high humidity for short time (2) Plantlet is placed in favourable conditions for long time (3) Plantlet is transfer in new fresh medium (4) Plantlet is placed in highlight intensity <p>117. In Mung bean, resistance to yellow mosaic virus and powdery mildew were</p> <ol style="list-style-type: none"> (1) Induced by mutations (2) Introduced by classical breeding (3) Carried out by mass selection (4) Transferred from a wild species <p>118. Pusa sadabahar is an improved variety of for</p> <ol style="list-style-type: none"> (1) Chilli, Bacterial blight (2) Brassica, Black rot (3) Chilli, chilly mosaic virus and leaf curl (4) Wheat, leaf and stripe rust | <p>119. Maize hybrids commonly have amino acid</p> <ol style="list-style-type: none"> (1) Arginine and tryptophan (2) Lysine and cysteine (3) Cysteine and Methionine (4) Lysine and tryptophan <p>120. 'Jaya' and 'Ratna' are high yielding semi dwarf variety of rice. These varieties were developed in which country?</p> <ol style="list-style-type: none"> (1) Japan (2) Mexico (3) India (4) Philippines <p>121. The variety which formed the basis of green revolution in India is/are</p> <ol style="list-style-type: none"> (1) Sharbati sonora and pusa lerma of wheat & rice (2) Sharbati sonora and pusa lerma of rice (3) Sharbati sonora and pusa lerma of wheat (4) Sharbati sonora and pusa lerma of rice & wheat <p>122. Gyanogenic haploid plants are obtained through :-</p> <ol style="list-style-type: none"> (1) Anther culture (2) Ovule culture (3) Embryo culture (4) Shoot tip culture <p>123. Mutation breeding is useful in</p> <ol style="list-style-type: none"> (1) Self pollinating plants (2) Cross pollinating plants (3) Vegetatively propagated plants (4) More than one option is correct <p>124. Which of the following is crucial step to the success of plant breeding program?</p> <ol style="list-style-type: none"> (1) Collection of variability (2) Evaluation and selection of parents (3) Selection and testing of superior recombinant (4) Testing, release and commercialization of new cultivars <p>125. Three crops that contribute maximum to global food grain production are:</p> <ol style="list-style-type: none"> (1) Wheat, rice and maize (2) Wheat, rice and barley (3) Wheat, maize, sorghum (4) Rice, maize and sorghum |
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126. Classical plant breeding involves

- (1) Artificial selection only
- (2) Hybridization of pure lines followed by artificial selection
- (3) Hybridisation of heterozygous varieties followed by artificial selection
- (4) Artificial selection followed by hybridisation

127. Choose the incorrect match w.r.t the crop and its variety.

	Crop	Variety
(1)	Wheat	Himgiri
(2)	Chilli	Pusa sadabahar
(3)	Cow pea	Pusa komal
(4)	Cauliflower	Pusa gaurav

128. Genetic variability is the root of any breeding programme. It is available from

- (1) Only cultivated improved varieties
- (2) Only related species
- (3) Only wild relative
- (4) Wild relative, desi varieties, old local varieties and improved varieties

129. High aspartic acid, low nitrogen and sugar content leads to resistance against A in B.

	A	B
(1)	Stem	Maize
(2)	Bollworms	Cotton
(3)	Jassids	Maize
(4)	Leaf beetle	Wheat

130. *Saccharum barberi*

- (1) Was originally grown in South India
- (2) Was originally grown in North India
- (3) Had poor sugar content and yield
- (4) Both (2) and (3)

131. First man made cereal *Triticale* is

- (1) Octaploid
- (2) Pentaploid
- (3) Haploid
- (4) Heptaploid

132. Capacity to generate a whole plant from any cell/explant is termed as

- (1) Somatic hybridisation
- (2) Totipotency
- (3) Micropropagation
- (4) None of these

133. Which of the following techniques offer vast potential for manipulation of plants in-vitro to produce micropropagation and new varieties?

- (1) Down streaming process
- (2) Tissue culture
- (3) Parasexual Hybridisation
- (4) Both (2) and (3)

134. All of the following can induce the fusion between protoplast of two different plants of different varieties, except :-

- (1) Polyethylene glycol
- (2) Sodium nitrate
- (3) High frequency alternating electric field
- (4) Colchicine treatment

135. Guha and Maheshwari are related to

- (1) Embryo rescue
- (2) Anther culture
- (3) Somatic hybridisation
- (4) Cybrid culture

136. A structural and functional unit of nature where living organisms interact among themselves and also with the surrounding physical environment is ecosystem. Which of the following is not correct for ecosystem?

- (1) Term ecosystem was given by A.G. Tansley
- (2) It is self regulatory and self sustaining unit
- (3) It cannot maintain resilience
- (4) Regular input of energy is essential to maintain homeostasis

137. The structural component of ecosystem includes :-

- (1) Species composition
- (2) Stratification
- (3) Decomposition
- (4) Both (1) and (2)

138. Stratification means vertical distribution of :-
 (1) Individuals of different species in temperate needle like forest
 (2) Individual of same species
 (3) Trees, shrubs, and herbs
 (4) Bacteria, cyanobacteria and fungi
139. Species having fewer number of individuals but determine the ability of other species to exist and removal of which affect large number of species in an ecosystem is :-
 (1) Keystone species
 (2) Endemic link species
 (3) Edge species
 (4) Ecotone
140. The annual net primary productivity of the whole biosphere approximately is :-
 (1) 170 billion tons
 (2) 80 billion tons
 (3) 55 billion tons
 (4) 55 million tons
141. The rate of formation of new organic matter by consumers is known as :-
 (1) GPP
 (2) NCP
 (3) Secondary Productivity
 (4) Net production efficiency
142. Nutrient immobilisation is :-
 (1) Incorporation of nutrients in living micro organisms
 (2) Covalent linking of nutrients with one another
 (3) Loss of nutrients during fragmentation
 (4) Loss of nutrients during soil erosion
143. Based on the source of their nutrition of food organism occupy a specific position in the food chain, that is known as their :-
 (1) Trophic level (2) Niche
 (3) Habitat (4) Micro climate
144. Plant capture only% of the PAR.
 (1) 1-5 (2) 25
 (3) 2-10 (4) 1.6-5
145. Which one of the following pyramid would be inverted?
 (1) Pyramid of number in a grassland ecosystem
 (2) Pyramid of number in a tree ecosystem
 (3) Pyramid of Biomass in sea
 (4) More than one option is correct
146. The first biotic community which develops in a bare area is called pioneer community which is characterised by :-
 (i) High productivity
 (ii) Low productivity
 (iii) High multiplication
 (iv) Low multiplication
 (1) (i), (iii) (2) (i), (iv)
 (3) (ii), (iii) (4) (ii), (iv)
147. Primary succession takes comparatively longer time to achieve climax because :-
 (1) Soil is not formed in the beginning
 (2) Soil has only some propagules
 (3) Soil has excess of minerals
 (4) Soil has low humus content
148. Climax community differs from pioneer community in
 (1) Increase in structural complexity
 (2) Increase in species diversity
 (3) Niche specialisation
 (4) All are correct
149. What is pioneer community for hydrarch and xerarch succession respectively?
 (1) Zooplanktons and mosses
 (2) Lichens and phytoplanktons
 (3) Submerged plants and mosses
 (4) Phytoplanktons and lichens
150. How many of given ecosystem are natural and man-made ecosystem respectively?
Forest, Grassland, Crop field, River, Estuary, Fishery tank, Aquaria, Lakes
 (1) 5, 3 (2) 3, 5
 (3) 4, 4 (4) 6, 2

151. Select the correct sequences of stages in xerarch succession.
- (1) Lichens → Mosses → Perennial grasses → Annual grasses → Shrubs → Forests
 - (2) Lichens → Mosses → Annual grasses → Perennial grasses → Shrubs → Forests
 - (3) Mosses → Lichens → Perennial grasses → Annual grasses → Shrubs → Forests
 - (4) Mosses → Lichens → Shrubs → Annual Grasses → Perennial grasses → Shrubs → Forests
152. Arrange various stages of primary succession in a newly formed fresh water pond are given :-
- (A) Scrub stage
 - (B) Sub merged free floating plant stage
 - (C) Marsh meadow stage
 - (D) Reed swamp stage
 - (E) Phytoplanktons
 - (F) Submerged plant stage
 - (G) Forest
- (1) E, F, B, C, D, A, G
 - (2) E, B, F, D, C, A, G
 - (3) E, F, B, D, C, A, G
 - (4) E, B, C, D, B, A, G
153. A cell becomes turgid when placed in
- (1) isotonic solution
 - (2) hypertonic solution
 - (3) hypotonic solution
 - (4) none of the above
154. Two cells A and B are in contact with one another. The osmotic pressure of cell A is 12 atms and that of B 10 atms. The Turgor pressure of cell A is 8 atms and that of B is 4 atms. In which direction will the water move –
- (1) from cell A to cell B
 - (2) from cell B to cell A
 - (3) there will be no movement of water
 - (4) water can move either from cell A to cell B or from B to A
155. Pressure exerted in the tracheary elements of a xylem as a result of metabolic activity of roots which forces the water into xylem vessel and upwards into the stem for a certain height is
- (1) osmotic pressure
 - (2) turgor pressure
 - (3) root pressure
 - (4) atmospheric pressure
156. Which of the following plants open their stomata during night and close during day time –
- (1) fleshy xerophytes
 - (2) hydrophytes
 - (3) mesophytes
 - (4) none of the above
157. Guttation takes place through –
- (1) stomata
 - (2) wound
 - (3) water stomata
 - (4) lenticel
158. Transpiration is measured by
- (1) porometer
 - (2) potentiometer
 - (3) potometer
 - (4) sonometer
159. According to K^+H^+ exchange theory which substances enters in the guard cells –
- (1) K^+ in place of H^+
 - (2) Na^+ in place of K^+
 - (3) H^+ in place of oxygen
 - (4) oxalate in place of malate
160. According to Levitt which accumulates in vacuolar sap during opening of stomata –
- (1) K^+ ions
 - (2) malate ions
 - (3) H_2O
 - (4) K^+ malate and water
161. If PMA (Phenyl Mercuric Acetate) is sprayed upon the leaf –
- (1) transpiration increases
 - (2) transpiration is reduced
 - (3) guttation is increased
 - (4) bleeding is increased

162. Which is antitranspirant –
 (1) PMA (phenyl mercuric acetate)
 (2) abscisic acid (ABA)
 (3) DDT
 (4) (1) and (2)
163. In a girdled plant which dies first –
 (1) root
 (2) stem
 (3) root and stem together
 (4) both survive
164. The potential energy of water is referred to as
 (1) osmotic potential
 (2) water potential
 (3) solute potential
 (4) gravity potential
165. Which of the following equation represent water potential in plant ?
 (1) $\psi_w = \psi_s + \psi_p + \psi_g$ (2) $\psi_w = \psi_s + \psi_p - \psi_g$
 (3) $\psi_w = \psi_s - (\psi_s - \psi_g)$ (4) none of these
166. Water potential (ψ_w) inside plant cells is
 (1) positive (2) negative
 (3) zero (4) all of the above
167. In symplastic pathway the movement of water occurs from cell to cell through the
 (1) cell wall without crossing any membrane
 (2) tonoplast and cell wall
 (3) plasmodesmata
 (4) cell wall and plasmodesmata
168. Semipermeable membrane allows to pass through it
 (1) any molecule
 (2) only solute molecule
 (3) only macromolecules
 (4) only solvent molecules
169. Cohesion theory is the most accepted theory of water movement through plants. This theory was given by
 (1) S.R.Stout (2) Henry Dixon
 (3) D.R. Hoagland (4) Mendel
170. The walls of xylem vessels have strong affinity for water molecules due to the presence of
 (1) cellulose in xylem vessels
 (2) peptidoglycan
 (3) lignocellulose
 (4) phospholipid
171. The crops start wilting if excess fertilizers are added to the field. This is due to
 (1) imbibition (2) plasmolysis
 (3) osmosis (4) none of these
172. By raising the temperature the rate of transpiration can be
 (1) increased
 (2) decreased
 (3) no increase or decrease
 (4) no effect
173. For a fully turgid cell
 (1) O.P. = D.P.D. (2) O.P. = Zero
 (3) D.P.D. = Zero (4) T.P. = Zero
174. A protoplast is a cell :-
 (1) without cell wall
 (2) without plasma membrane
 (3) without nucleus
 (4) undergoing division
175. Main absorption of water occurs in the zone of
 (1) root cap
 (2) root hair zone
 (3) meristematic zone
 (4) cell elongation zone
176. Absorption of water is not affected by
 (1) water logging
 (2) soil temperature
 (3) soil air
 (4) soil water available in sufficient quantity
177. Lowering of temperature towards freezing leads to
 (1) increased absorption
 (2) decreased absorption
 (3) no effect
 (4) none of these
178. Root hair absorb water when
 (1) soil solution is isotonic
 (2) they respire rapidly
 (3) salt concentration of cell sap is high
 (4) salt concentration of soil is high
179. Plant cells submerged in distilled water will become
 (1) turgid (2) flaccid
 (3) plasmolysed (4) impermeable
180. The loss of water in the form of water drops is called
 (1) transpiration (2) guttation
 (3) respiration (4) root pressure